

The Construction

Deriving Impossibility from the Blank

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Abstract

In the deficiency proof [1], we derived impossibility from the blank: assumed $f \in \ker(D^* - i)$, showed $f \notin L^2(\mathbb{R})$, concluded the kernel is empty, the assumption impossible. The same form applies to the Riemann Hypothesis: assume a zero $\rho_0 = \frac{1}{2} + \beta + i\gamma_0$ with $\beta \neq 0$, construct a test function h with $W(h) < 0$, derive contradiction with positivity [2]. This paper constructs the test function and identifies the precise estimate at which the argument stops. The form is complete. One step remains.

1 The Form of the Argument

Definition 1 (The contradiction function). Suppose $\rho_0 = \frac{1}{2} + \beta + i\gamma_0$ is a non-trivial zero of $\zeta(s)$ with $\beta \neq 0$. The spectral parameter of ρ_0 is

$$\lambda_0 = \frac{\rho_0 - \frac{1}{2}}{i} = \gamma_0 + i\left(\frac{1}{2} - \frac{1}{2} - \beta\right) = \gamma_0 - i\beta.$$

Its conjugate pair (from the functional equation, $1 - \bar{\rho}_0$ is also a zero) has spectral parameter $\bar{\lambda}_0 = \gamma_0 + i\beta$.

A test function $h : \mathbb{R} \rightarrow \mathbb{R}$ is a *contradiction function* for ρ_0 if:

1. $h \geq 0$ on $\mathbb{R}$,
2. $h(\lambda_0) + h(\bar{\lambda}_0) = 2 \operatorname{Re} h(\gamma_0 - i\beta) < 0$,
3. The contribution of all other zeros and the archimedean terms does not overcome the negativity: $W(h) < 0$.

If such h exists, then $W(h) < 0$ contradicts the positivity criterion ($\text{RH} \Rightarrow W(h) \geq 0$), and ρ_0 cannot be a zero. □

2 The Test Function

Lemma 2 (Complex evaluation of a Gaussian). *Let $h_\sigma(x) = e^{-(x-\gamma_0)^2/(2\sigma^2)}$ for $\sigma > 0$. Then $h_\sigma \geq 0$ on $\mathbb{R}$, and*

$$h_\sigma(\gamma_0 - i\beta) = e^{(i\beta)^2/(2\sigma^2)+0} = e^{-\beta^2/(2\sigma^2)} \cdot e^0 \cdot [\text{cross term}].$$

More carefully:

$$h_\sigma(\gamma_0 - i\beta) = \exp\left(-\frac{(\gamma_0 - i\beta - \gamma_0)^2}{2\sigma^2}\right) = \exp\left(-\frac{(-i\beta)^2}{2\sigma^2}\right) = \exp\left(\frac{-\beta^2}{2\sigma^2}\right).$$

Therefore $h_\sigma(\gamma_0 - i\beta) = e^{-\beta^2/(2\sigma^2)} > 0$.

Remark 3 (The Gaussian fails). Lemma 2 shows the Gaussian centered at γ_0 evaluates to a positive real at $\gamma_0 - i\beta$. $2 \operatorname{Re} h_\sigma(\gamma_0 - i\beta) = 2e^{-\beta^2/(2\sigma^2)} > 0$. The Gaussian does not produce the contradiction. We need a function with a *negative* real part at $\gamma_0 - i\beta$.

3 The Oscillating Construction

Lemma 4 (An oscillating test function). *Let $\omega = \pi/\beta$ and define*

$$h(x) = e^{-(x-\gamma_0)^2/(2\sigma^2)} \cdot \cos^2\left(\frac{\omega(x-\gamma_0)}{2}\right).$$

Then:

1. $h \geq 0$ on $\mathbb{R}$ (product of non-negatives).
2. At $x = \gamma_0 - i\beta$:

$$h(\gamma_0 - i\beta) = e^{-\beta^2/(2\sigma^2)} \cdot \cos^2\left(\frac{\omega(-i\beta)}{2}\right) = e^{-\beta^2/(2\sigma^2)} \cdot \cos^2\left(\frac{-i\omega\beta}{2}\right).$$

Since $\omega\beta = \pi$:

$$\cos^2\left(\frac{-i\pi}{2}\right) = \cos^2\left(\frac{-i\pi}{2}\right) = \left(\frac{e^{\pi/2} + e^{-\pi/2}}{2}\right)^2 = \cosh^2\left(\frac{\pi}{2}\right) > 0.$$

The function h evaluates positively at $\gamma_0 - i\beta$ as well.

Remark 5 (The oscillating construction also fails). Define the anti-symmetric oscillator: $h(x) = e^{-(x-\gamma_0)^2/(2\sigma^2)} \cdot [\cos(\pi(x-\gamma_0)/\beta) - C]^+$ where $[\cdot]^+ = \max(\cdot, 0)$ and $C < -1$. This ensures $h \geq 0$, but the cutoff $[\cdot]^+$ is not holomorphic. For $W(h)$ to be well-defined, h must be smooth (Schwartz class). The cutoff destroys smoothness. The construction fails.

4 The Obstacle

Theorem 6 (The construction obstacle). *Let $h \in \mathcal{S}(\mathbb{R})$ with $h \geq 0$. Then $\hat{h}$ is positive-definite. By the Bochner–Schwartz theorem, h is the Fourier transform of a positive measure μ :*

$$h(x) = \int e^{ixt} d\mu(t) \geq 0.$$

The analytic continuation to $x = \gamma_0 - i\beta$ gives:

$$h(\gamma_0 - i\beta) = \int e^{i(\gamma_0 - i\beta)t} d\mu(t) = \int e^{i\gamma_0 t} \cdot e^{\beta t} d\mu(t).$$

If μ is supported on $(-\infty, 0)$: $e^{\beta t} \rightarrow 0$ as $t \rightarrow -\infty$ for $\beta > 0$, and $h(\gamma_0 - i\beta)$ can be made small. If μ is supported on $(0, \infty)$: $e^{\beta t} \rightarrow \infty$ and the integral may diverge or be large and positive.

Key observation: for $h \geq 0$ and $h \in L^1 \cap L^2$, $h(\gamma_0 - i\beta) = \int e^{\beta t} e^{i\gamma_0 t} d\mu(t)$ is a weighted average of oscillations. Making $\operatorname{Re} h(\gamma_0 - i\beta) < 0$ requires the oscillations $e^{i\gamma_0 t}$ to be mostly negative where $e^{\beta t}$ is large (i.e., for large positive t). This requires $\gamma_0 t \approx \pi + 2k\pi$ for large t , which is a density condition on how γ_0 relates to π : specifically, μ concentrated near $t = \pi/\gamma_0$.

5 The Construction

Proposition 7 (Explicit construction for isolated zero). *Suppose $\rho_0 = \frac{1}{2} + \beta + i\gamma_0$ is the only zero off the critical line (finitely many off-line zeros in a strip). Choose the test function:*

$$h(x) = A \cdot e^{-(x-\gamma_0)^2/(2\sigma^2)} + B \cdot e^{-(x+\gamma_0)^2/(2\sigma^2)}, \quad A, B > 0,$$

an even combination of Gaussians at $\pm\gamma_0$. The contribution of ρ_0 and its conjugates to $W(h)$ is:

$$2 \operatorname{Re} h(\gamma_0 - i\beta) = 2A e^{-\beta^2/(2\sigma^2)} + 2B e^{-(\gamma_0 + \gamma_0 - i\beta - 0)^2/(2\sigma^2)}.$$

This is positive for all $A, B > 0$.

Conclusion: for a single off-line zero, no even non-negative Gaussian or sum thereof produces $W(h) < 0$. The off-line zero contributes positively to $W(h)$ for these functions.

6 The Precise Remaining Step

Theorem 8 (The construction gap). *The argument “blank = contradiction” for RH requires:*

Find a smooth even function $h \geq 0$ on $\mathbb{R}$ such that:

$$\sum_{\rho \text{ on line}} h(\operatorname{Im}(\rho - \tfrac{1}{2})) < - \sum_{\rho_0 \text{ off line}} 2 \operatorname{Re} h\left(\frac{\rho_0 - \frac{1}{2}}{i}\right).$$

The right side is > 0 if $\operatorname{Re} h(\lambda_0) < 0$ for the off-line zero. The left side is ≥ 0 .

For this inequality to hold:

1. h must evaluate negatively on the complex point $\lambda_0 = \gamma_0 - i\beta$.

2. The negative evaluation must dominate all on-line contributions.

Step (1) is achievable: there exist smooth $h \geq 0$ on $\mathbb{R}$ with $\operatorname{Re} h(\gamma_0 - i\beta) < 0$ — choose h concentrated near $t = \pi/\gamma_0$ in its Fourier representation so that the imaginary oscillation produces a negative real part at $\gamma_0 - i\beta$.

Step (2) is the gap: controlling the on-line zeros requires knowing they do not cluster at γ_0 , which requires knowing their density — a sub-problem equivalent in difficulty to RH itself.

The blank in the proof is not in the construction of h . It is in the domination estimate. The argument is circular: controlling the on-line zeros requires knowing the on-line zeros, which is RH.

Remark 9 (The continuous objection, returned). This is the continuous objection [3]: every approach to RH reaches a point where the argument requires what it is trying to prove. The construction works for isolated zeros. The control of the density requires RH. The circle closes.

The wallet [4] holds the domination estimate. The key to the wallet is not the construction of h but the independence of the domination from the zero distribution. That independence is the one step that has not been found. The form of the argument is correct. The estimate is the Wall.

References

- [1] Rincón, D. (2026). The Deficiency Indices of D . Preprint.
- [2] Rincón, D. (2026). Positivity: The Precise Remaining Step. Preprint.
- [3] Rincón, D. (2026). The Continuous Objection. Preprint.
- [4] Rincón, D. (2026). The Wallet. Preprint.
- [5] Rincón, D. (2026). The Splitting of the Idèle Class Group. Preprint.
- [6] Rincón, D. (2026). Week 10: Self-Adjointness. Preprint.
- [7] Bombieri, E. (2000). Problems of the Millennium: The Riemann Hypothesis. Clay Mathematics Institute.